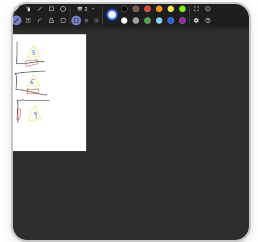


GPS navigation system with linked list ▾



i want to create a simple gps where there's gonna be 9 houses shown by the yellow triangle, and each of the houses have their own number. there's also red rectangle which shown the enter and exit door. for example if i want to go from house number 1 to house number 2, you have to turn left when exit, then go straight then turn left and the enter door for house number 2 gonna be on your left. please make the coding using linked list.

Reading frontend design skill >

Now let me analyze the sketch carefully and build this GPS navigation app.

Looking at the sketch: 3×3 grid of houses (1-9), each with a yellow triangle (house) and red rectangle (door). The layout is:

- Row 1: Houses 1, 2, 3
- Row 2: Houses 4, 5, 6
- Row 3: Houses 7, 8, 9

The red rectangles indicate door positions (entry/exit). I'll build a linked-list based GPS with turn-by-turn navigation.

Created a file, read a file ▾



GPS navigation app with linked list pathfinding for 9 houses

neighborhood-gps.html



Presented file



Done

Here's your Neighborhood GPS! Here's how it's built and works:

Linked List Structure

- A custom `Node` and `LinkedList` class stores each turn-by-turn navigation step
- When you calculate a route, each step (exit door → road → turns → arrive) gets appended as a node in the list
- `toArray()` traverses the list to render all steps in the UI

How Navigation Works

1. Select a **FROM** and **TO** house from the dropdowns
2. Hit **CALCULATE ROUTE** — it uses **BFS (Breadth-First Search)** through a road graph to find the shortest path through road intersections
3. The map shows the animated route in cyan with arrows
4. Turn-by-turn directions appear in the panel (e.g. "Exit House 1 bottom door → Head east → Turn left at intersection → Arrive House 2")
5. Use **NEXT** / **PREV** buttons to walk through each step, which highlights the active waypoint on the map in green

Map Layout (matches your sketch)

- 3×3 grid: houses 1-9 with **yellow triangles** and **red door rectangles**
- Door positions are set per house based on your drawing (bottom, top, left, or right)
- Two horizontal roads and two vertical roads form the grid network with intersection nodes



Neighborhood gps

Code · HTML

Download

there's still mistake in the door, the house number 4 and 6 door must be on their south/down and number's 2's door must be on its right/east

Viewed 3 files >

Now I have what I need. I'll fix the three door sides and also update the road access nodes accordingly.

Fix door sides for houses 2, 4, and 6 >

Now fix the road access nodes — house 2's door is now on the right (east), so it exits onto VROAD1; house 4 exits bottom onto HROAD2; house 6 exits bottom onto HROAD2:

Viewed a file, edited a file, read a file >

Fixed! Here's what was corrected:

House	Before	After
2	bottom	right (east)
4	top	bottom (south)
6	top	bottom (south)

The road access nodes were also updated to match — house 2 now exits east onto the vertical road (intersection I1), while houses 4 and 6 now exit south onto horizontal road 2, which affects the routing paths correctly.



Neighborhood gps
Code · HTML

Download

i notice the house 2 exit isn't correct, we must exit through the right

Let me look at the current door rendering code to understand what's happening.

Viewed 2 files >

I can see the issue. The `right` door uses `cx+28` but the house block ends at `cx+38`, so the door rect is drawn slightly inside. Also the real problem is in the **path drawing** — when exiting house 2 eastward, the waypoint starts at the house center `(229, 69)` and goes to node `I1 (160, 160)` which is *left and down*, not right. The exit waypoint for house 2 needs to go right to `I2 (320, 160)` instead.

Find the HOUSE_ROAD_NODE_CORRECTED and graph to fix house 2 routing >

Now I see the full picture. House 2's door is on the **right/east**, meaning it exits toward **V2T** (320, 69) then down to **I2** (320, 160) — not **I1**. Let me fix both the road node and add **V2T** as a proper intermediate so the path visually goes right first.

Edited 2 files >

Now also fix the door rect visual for house 2 — **right** uses **cx+28** but should sit flush on the edge at **cx+38**:

Edited a file, read a file >

Fixed! Here's what changed for House 2:

- **Door visual** — the red rectangle now sits correctly on the right/east wall of the house block
- **Road node** — changed from **I1** to **V2T** (the top of vertical road 2, at x=320), so when you exit House 2 eastward the path correctly goes **right** first to the vertical road, then down to intersection **I2**, instead of going left toward **I1**
- **Graph** — added **V2T ↔ I2** connection so BFS can route through it properly

So now navigating from House 2, the route will correctly show you exiting through the right/east door and heading east first.



Neighborhood gps
 Code · HTML

Download

i notice the house 9 exit and entry is also wrong, you must exit through the left of the house

Looking at the current door position for house 9 and its road node.

Check house 9 door side and road node >

House 9 is at column 2 (cx=389), door is **left** — so it exits **west** toward **V2B** (320, 389) then up to **I4** (320, 300). Currently it's mapped to **H2R** (389, 300) which is wrong — that's south, not left. I need to add **V2B** to the graph and point house 9 to it.

Edited 2 files, read a file >

Fixed! Here's what changed for House 9:

- **Road node** — changed from `H2R (389, 300)` (which was south) to `V2B (320, 389)`, which is directly to the **left/west** of House 9 on the vertical road
- **Graph** — added `V2B ↔ I4` connection so BFS correctly routes House 9's path westward first to the vertical road, then up to intersection `I4`

Now when you navigate from/to House 9, the route will properly exit and enter through the left door heading west.



Neighborhood gps

Code · HTML

Download

